GAME DEVELOPMENT FOR SOCIAL GOOD

Common Ground

A city-building game about making a city fair, healthy and sustainable

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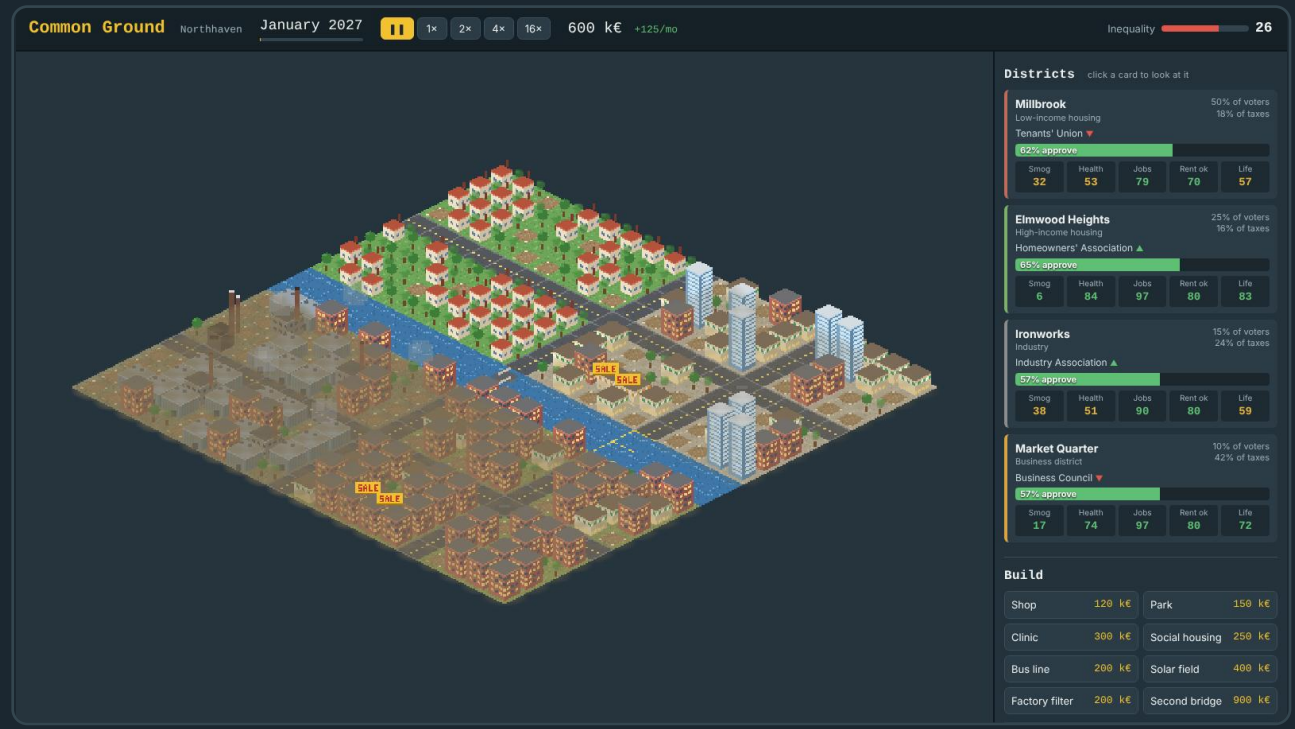


You are the mayor of Northhaven

Build housing, parks, clinics, bus stations and clean energy on an isometric map.

Manage the budget, taxes and policies for each of the four districts, month by month.

Answer to voters: every December the residents decide whether you stay mayor.



THE CORE TENSION

Who pays and who votes are different people

District	Who lives there	Voters	Taxes
Millbrook	Low-income housing	50%	18%
Elmwood Heights	High-income housing	25%	16%
Ironworks	Factories, coal plant	15%	24%
Market Quarter	Shops and offices	10%	42%

Building in the rich districts brings money back fast. Building in Millbrook helps the most people and wins votes, but pays back slowly.

SOCIAL GOOD THEME

A fair, healthy and sustainable city

Inequality SDG 10	**Air pollution and health** SDG 3, 11.6	**Clean energy** SDG 7, 13	**Affordable housing** SDG 11.1
Public transport SDG 11.2	**Green space, clean river** SDG 6, 11.7	**Jobs and local shops** SDG 8	**Underneath all: short-term thinking**

Main goals: **SDG 11** Sustainable Cities and **SDG 10** Reduced Inequalities. Players learn to ask: who benefits, who pays, and what does it cost the environment?

IMPLEMENTED

What works in the prototype

City map and building

Isometric map with a river and 4 districts; 7 building types plus factory filters.

Time and budget

1 month = 1 minute, from pause to 16× speed; taxes in, upkeep out, a debt limit.

Taxes and policies

Residential and business tax per district; a rent cap for Millbrook.

Factions and fairness

4 factions with approval bars; an inequality meter compares the districts.

Events, election

3 events, a December vote by district and a report card.

Play-data logging

Every month is recorded and can be downloaded as CSV for evaluation.

Problems show up on the map



Smog

Factory smoke drifts east over Millbrook, and residents' health slowly falls.

SALE, then FOR RENT

Shops without customers struggle, then close: fewer jobs, less tax.

Protests and moving trucks

Angry factions protest at the town hall; rising rents push families out.

Every choice has winners and losers

Yearly goal	Tenants	Homeowners	Industry	Business
Cut factory smoke by 25%	+8	+3	−10	0
Open a bus line from the south	+6	−2	+2	+5
Lower business tax to 8%	−5	+3	+8	+8
Build social housing	+8	−5	+2	−2



Events force choices too

Take 400 k€ for luxury flats, and rents in Millbrook start rising. Or refuse and keep the riverbank public.

Common Ground · 9 · approval change when a goal is chosen

December: the city votes



Election night: Re-elected!

You won 62.6% of the vote against The Builder.

District	Approval	Voters	Voted for you
Millbrook	72%	3,200	72%
Elmwood Heights	53%	1,181	47%
Ironworks	64%	819	56%
Market Quarter	53%	468	50%

Your promises

- ❌ Broken: Cut factory smoke by 25%
- ✅ Kept: Open a bus line from the south bank
- ✅ Kept: Build social housing

City report card

Fairness between districts	gap 21.7 (start 25.9)	D
Air quality in Millbrook	81 / 100	B
Affordable housing in Millbrook	69 / 100	B
Jobs in Millbrook	85% employed	B
Local shops	34 open, 0 struggling, 0 closed	A
City budget	326 k€	B

Ending: "Common Ground". You stayed in office and the gap between districts shrank.

Prototype: the full game continues with a new year here.

Download play data (CSV) | Play again

Test runs of one year

Strategy	Vote share
Do nothing	31%, voted out
Spend only on rich districts	35%, voted out
Balanced plan	58–75%, re-elected

The balanced plan still angers industry. The report card grades fairness, air, housing, jobs, shops and budget, so winning by neglect is visibly a worse result.

PLATFORM AND TOOLS

How it is built

Phaser 3 scene
Isometric map, camera, clicks, smoke and signs

HTML and CSS interface
District cards, build menu, taxes, pop-ups

Simulation in plain JavaScript
Taxes, pollution, jobs, shops, rents, approval, votes

One data file
Every number in the game, so balancing needs no code changes

FINAL VERSION · DUE 1 DECEMBER 2026

What comes next

Not yet implemented

- Real pixel-art sprites and sound
- All three rival candidates
- Several years instead of one
- River clean-up and the riverfront choice
- More events and a short tutorial

AI USAGE

How AI was used

Task	What Claude (Anthropic) did
Design	UI design for prototype
Code	Generated the game code in Phaser 3 and helped with javascript integration
Testing	Automated one-year test runs in a browser to balance the numbers

LIVE DEMO

One year in Northhaven

Can you stay mayor and close the gap?

Space: pause · 1 to 4: speed · N: skip a month · drag to move · mouse wheel to zoom